

Agent Acceleration in a Stable Environment

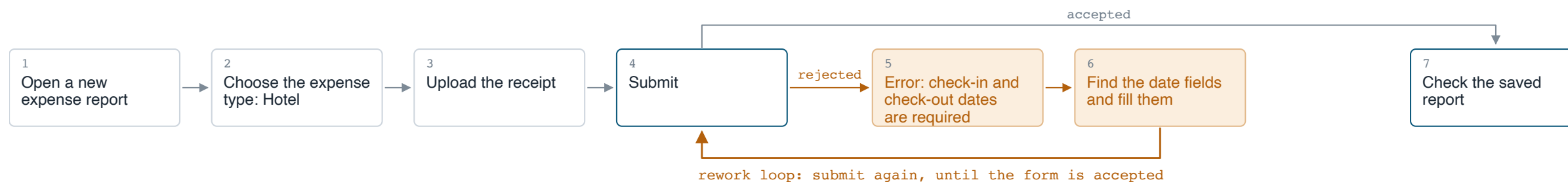
Learning from repeated tasks to make later ones faster and cheaper

October 2026

Example: a web agent that files expense reports

Our running example: a web agent files expense reports; each task is a new report in the same system. Not knowing one rule of the form, that a hotel expense needs check-in and check-out dates, sends the agent into a rework loop.

ONE HOTEL REPORT, BEFORE THE AGENT KNOWS THE RULE



WITHOUT LEARNING

The loop is paid again on every hotel report

- Its extra model calls and seconds are part of $\bar{v}(m)$, for each of the n reports

LEARNING THE RULE ONCE

The rule is learned once, at cost C_{learn} ; later reports skip the loop

- $\bar{v}(m') < \bar{v}(m)$; it pays off after n^* reports (page 1)

IN THIS EXAMPLE

- **Environment:** the expense system, its forms, the fields each expense type requires, the actions it allows
- **Run record:** the steps above, with the error message, and the time and cost of each step

WHAT LEARNING MAY CHANGE HERE (MODEL WEIGHTS FIXED)

- **Prompt:** an instruction read on every call: fill the check-in and check-out dates before submitting a hotel expense
- **Memory:** a note read only when relevant: Hotel → check-in and check-out dates required
- **Skill or tool:** a procedure the model calls when it needs it: instructions plus a script that fills every field a hotel expense requires
- **Control code:** the harness's own logic, run whatever the model decides: check the required fields before every submit

SCOPE → this example: expense reports on a website → any workflow repeated in one software environment → websites · desktop applications · mobile apps · tool APIs

Five difficulties, grouped by the cost they affect

COST PER TASK AFTER LEARNING · $\bar{v}(m')$

1 · Fewer steps do not always save time

BEFORE

AFTER

one step

extra reading and thinking in each step

length = seconds per task · per step 20.1 → 38.7 s (calc.)

Each step got longer: judge a report by its seconds, not its steps

Drawn from WebCoach (Liu et al., 2025). Also: ReasoningBank (Ouyang et al., 2026), fewer steps, more tokens

2 · Whether a check is needed is not known in advance

Choose type: Hotel

check the form

fill dates

submit

✓

schematic

one step more, always right

skip: as last time

submit

✓ rule still holds

✗ it changed: rework

Skipping a check saves a step only while the rule still holds

Seen in: DRAFT (Qu et al., 2025), learns tool conditions by trying · SKILL.nb (Hattami et al., 2026), checks before reuse

3 · A trade-off between generalization and cost per use

generalizes ↑

cheap and general?

note in every prompt

note in memory

skill: steps + script

replayed script

schematic

cost per use →

Seen in: Metis (Dai et al., 2026), keeps both text and code · ActionEngine (Zhong et al., 2026), turns a site map into programs

COST OF LEARNING · C_{learn}

4 · Finding the cause behind an error message

Error: over the meal limit

possible causes, of different kinds, checked one by one

reasoning wrong type?

rule limit changed?

interaction wrong field?

tool amount misread?

...?

the cause may be found late, or not at all

cost of checking so far →

Each possible cause costs a check; the right one may come late, or never

Seen in: HarnessFix (Chen et al., 2026), finds the faulty step · ESPO (L. Liu et al., 2026), sorts errors first

5 · Learning and testing can cost more than they save

total cost ↑

learning and testing are paid first

break-even n^*

break-even, more testing

repair after the site changes

no learning more testing with learning

schematic

number of reports filed →

Learning pays off only if enough reports share its cost

Seen in: AI Agents That Matter (Kapoor et al., 2025), late break-even · SICA (Robeyns et al., 2025), large bill, small saving

REFERENCES

• Chen, M., Wang, J., Liu, Z., et al. (2026). From Failed Trajectories to Reliable LLM Agents: Diagnosing and Repairing Harness Flaws. *arXiv preprint*.

• Dai, Z., He, S., Li, H., et al. (2026). Metis: Bridging Text and Code Memory for Self-Evolving Agents. *arXiv preprint*.

• Hattami, A. E., Chapados, N., & Pal, C. (2026). SKILL.nb: Selective Formalization and Gated Execution for Durable Agent Workflows. *FAGEN@ICML 2026 Poster (ICML 2026 workshop, venueid ICML.cc/2026/Workshop/FAGEN)*.

• Kapoor, S., Stroebel, B., Siegel, Z. S., et al. (2025). AI Agents That Matter. *Transactions on Machine Learning Research*.

• Liu, G., Geng, S., Li, S., et al. (2025). WebCoach: Self-Evolving Web Agents with Cross-Session Memory Guidance. *ICLR 2026 Workshop on Memory for LLM-Based Agentic Systems (MemAgents)*.

• Liu, L., Tang, P., Singh, K. Y., et al. (2026). ESPO: Error-Structured Prompt Optimization via Diagnose, Diversify, and Stabilize. *EMNLP 2026*.

• Ouyang, S., Yan, J., Hsu, I. H., et al. (2026). ReasoningBank: Scaling Agent Self-Evolving with Reasoning Memory. *ICLR 2026*.

• Qu, C., Dai, S., Wei, X., et al. (2025). From Exploration to Mastery: Enabling LLMs to Master Tools via Self-Driven Interactions. *ICLR 2025 (International Conference on Learning Representations 2025)*.

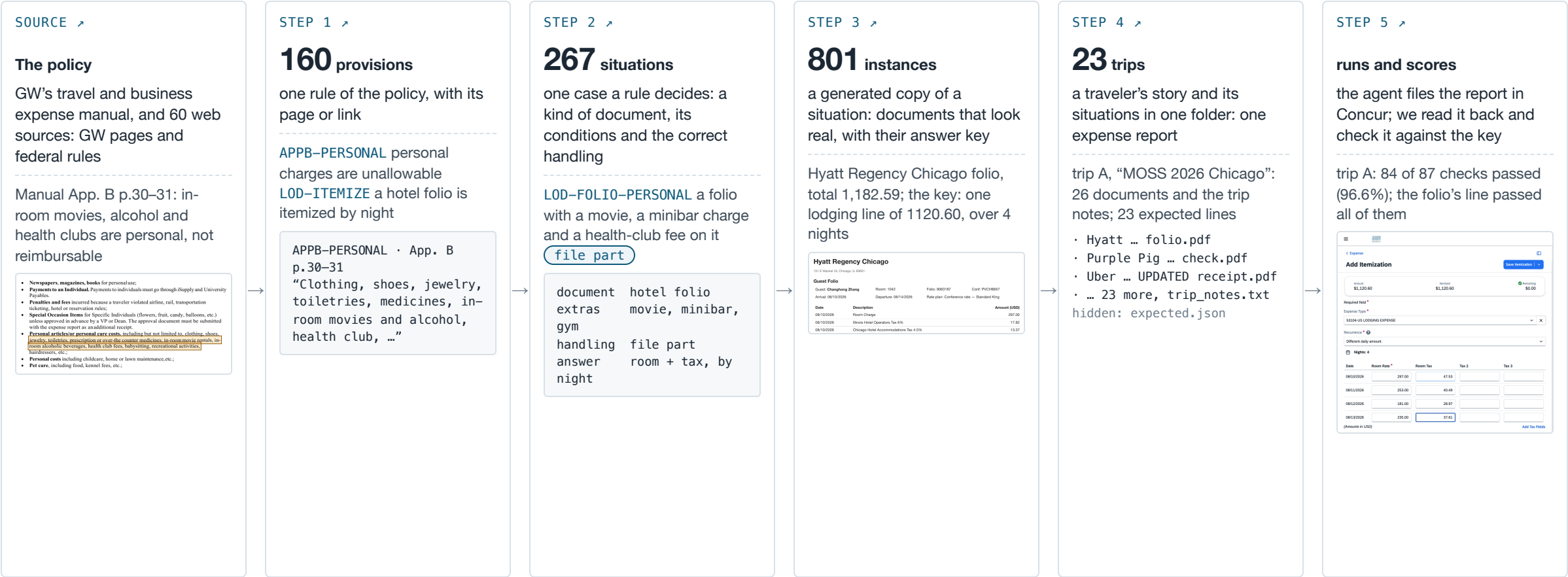
• Robeyns, M., Szummer, M., & Aitchison, L. (2025). A Self-Improving Coding Agent. *ICLR 2025 Workshop 'Self-Improving Foundation Models Without Human Supervision'*.

• Zhong, H., Faisal, F., França, L., et al. (2026). ActionEngine: From Reactive to Programmatic Web Agents via State Machine Memory. *arXiv preprint*.

From the expense policy to a scored report, in five steps

We break GW’s expense policy into rules, write each case a rule decides as a situation, generate documents and an answer key for it, bundle situations into expense reports, and score each report the agent files against its key.

FOLLOWED THROUGH THIS SECTION: ONE HOTEL FOLIO WITH PERSONAL CHARGES ON IT, FROM TRIP A (CHICAGO)



Provisions: the policy as 160 rules, each with its page or link

A provision is one rule of the policy, recorded with its page or link; the 149 rules an agent can act on while filing are each tested by at least one situation.

WHAT THE MANUAL SAYS · APPENDIX B, P.30–31

- **Newspapers, magazines, books** for personal use;
- **Payments to an Individual.** Payments to individuals must go through iSupply and University Payables.
- **Penalties and fees** incurred because a traveler violated airline, rail, transportation ticketing, hotel or reservation rules;
- **Special Occasion Items** for Specific Individuals (flowers, fruit, candy, balloons, etc.) unless approved in advance by a VP or Dean. The approval document must be submitted with the expense report as an additional receipt.
- **Personal articles/or personal care costs.** including but not limited to, clothing, shoes, jewelry, toiletries, prescription or over-the counter medicines, in-room movie rentals, in-room alcoholic beverages, health club fees, babysitting, recreational activities, hairdressers, etc.;
- **Personal costs** including childcare, home or lawn maintenance, etc.;
- **Pet care**, including food, kennel fees, etc.;

highlighted: the personal charges the manual excludes

RECORDED AS A PROVISION
APPB–PERSONAL · App. B p.30–31
“Clothing, shoes, jewelry, toiletries, medicines, in-room movies and alcohol, health club, babysitting, recreation, hairdressers, haircuts, personal care”

TESTED BY 14 SITUATIONS
LOD–FOLIO–PERSONAL, LOD–FOLIO–INROOM–ALCOHOL, GT–PERSONAL–DETOUR, OTH–CLOTHING, ...

READ BY THE AGENT
distilled into its rulebook, rules.md, under “Skip these — unallowable”

KIND 1 · WHERE A PROVISION COMES FROM

- 83 the manual** (rev. 2024-02-21)
[LOD–ITEMIZE](#) p.12–13: a hotel folio is itemized by night, room rate and tax; personal charges on it are left out
- 77 60 web sources:** GW pages and federal rules, for expenses outside travel
[NC–SELF–PAID](#): subscriptions, supplies, cloud services and dues paid out of pocket are filed, with a notice that GW prefers its P-Card or iBuy+

KIND 2 · CAN AN AGENT ACT ON IT WHILE FILING?

- 149 yes:** it decides how a document is filed
[APPB–PERSONAL](#), on the left
- 11 no:** it acts before or after filing
[ADVANCE](#) p.18–19: a travel advance is reconciled against the report, not filed from a receipt

Listing the 11 keeps “the whole policy is covered” honest: it means every rule an agent can act on, not only the convenient ones.

Situation: a document, its conditions and its correct handling

A situation is one case the policy decides, apart from any trip; the same document under other conditions is another situation.

ONE SITUATION, AS WE WROTE IT

SITUATION	LOD-FOLIO-PERSONAL family: lodging
DOCUMENT	a hotel folio with an in-room movie, a minibar charge and a health-club fee on it
CONDITIONS	a trip to a US city; any traveler; any funding
HANDLING	file part the room and its tax only, itemized by night
ANSWER KEY	one lodging line, night by night; the folio's total must not appear as a line
BASIS	manual p.12–13 and App. B p.30–31; one of our own hotel folios
FORM WORK	complex: the nights are entered on a second screen

Hyatt Regency Chicago		
151 E Wacker Dr, Chicago, IL 60601		
Guest Folio		
Guest: Changhong Zhang	Room: 1042	Folio: 9063187
Arrival: 08/10/2026	Departure: 08/14/2026	Conf: P/CH88X7
Rate plan: Conference rate — Standard King		
Date	Description	Amount (USD)
08/10/2026	Room Charge	297.00
08/10/2026	Illinois Hotel Operators Tax 6%	17.82
08/10/2026	Chicago Hotel Accommodations Tax 4.5%	13.37
08/10/2026	Cook County Hotel Tax 1%	2.97
08/10/2026	MPEA Hotel Tax 2.5%	7.43
08/10/2026	Sports Facilities Tax 2%	5.94
08/11/2026	Room Charge	253.00
08/11/2026	Illinois Hotel Operators Tax 6%	15.18
08/11/2026	Chicago Hotel Accommodations Tax 4.5%	11.39
08/11/2026	Cook County Hotel Tax 1%	2.53
08/11/2026	MPEA Hotel Tax 2.5%	6.33
08/11/2026	Sports Facilities Tax 2%	5.06
08/11/2026	In-Room Movie	14.99
08/12/2026	Room Charge	181.00
08/12/2026	Illinois Hotel Operators Tax 6%	10.86
08/12/2026	Chicago Hotel Accommodations Tax 4.5%	8.15
08/12/2026	Cook County Hotel Tax 1%	1.81
08/12/2026	MPEA Hotel Tax 2.5%	4.53
08/12/2026	Sports Facilities Tax 2%	3.62
08/12/2026	Minibar	22.00
08/13/2026	Room Charge	235.00
08/13/2026	Illinois Hotel Operators Tax 6%	14.10
08/13/2026	Chicago Hotel Accommodations Tax 4.5%	10.58
08/13/2026	Cook County Hotel Tax 1%	2.35
08/13/2026	MPEA Hotel Tax 2.5%	5.88
08/13/2026	Sports Facilities Tax 2%	4.70
08/13/2026	Health Club / Fitness Center	25.00
Total		1182.99 USD
Payment — Visa XXXX4417		-1182.99 USD
Balance		0.00 USD

one generated copy (trip A): the three personal charges and the total are highlighted

THE SAME DOCUMENT, OTHER CONDITIONS

The Westin Copley Place, Boston		
10 Huntington Ave, Boston, MA 02116 -		
Guest Folio		
Guest: Changhong Zhang	Room: 1846	Folio: 4713770
Arrival: 08/24/2026	Departure: 08/27/2026	Conf: N8KR8QRM
Rate plan: Conference rate — Standard King		
Date	Description	Amount (USD)

a folio with a high-speed internet charge on it

SITUATION	LOD-FOLIO-INTERNET
CONDITION	paid from department funds
HANDLING	file
ANSWER KEY	one lodging line, the internet included

SITUATION	LOD-FOLIO-INTERNET-GRANT
CONDITION	paid from a federal research grant
HANDLING	file + flag
ANSWER KEY	one lodging line, and a flag: internet is unallowable on the grant

267 situations: 169 in travel, 98 outside it. None is tied to a trip: each can be generated for any city, dates, traveler and funding its conditions allow.

Five ways a document can be handled

Every situation has one correct handling; 196 of the 267 ask for more than copying the receipt onto an expense line, and these judgment calls are where an agent goes wrong.

file

71 situations

one expense line, as the receipt reads

UNITED

Your Flight Receipt — CHANGHONG ZHANG — 24AUG26

Confirmation Number
XN8PE2

Passenger Info
Name: CHANGHONG ZHANG

FLIGHT	DATE	DEPART	ARRIVE
UNITED 1664 Economy (Y)	MON, 24AUG	RONALD REAGAN WASHINGTON NATIONAL AIRPORT 07:00AM	BOSTON LOGAN INTERNATIONAL AIRPORT 08:31AM

AIR-COACH-DIRECT a coach ticket bought from the airline → one airfare line

file + flag

97 situations

one line, and a note for a person to decide before the report is submitted

NEDS Society

101 Federal St, Suite 1000, Boston, MA 02110
(617) 555-5142
https://nedsociety.org
registration@nedsociety.org

THANK YOU!

Receipt Info
Date: 8/4/2026
Receipt Number: 3887364828
Customer: Changhong Zhang
Total: \$190.00

QUANTITY	LINE DESCRIPTION	LIST PRICE	SALE PRICE	TOTAL
1	Virtual Workshop on Thoughtworthy ML — online registration	\$190.00	\$190.00	\$190.00
			SUBSTOTAL:	\$190.00
			TOTAL:	\$190.00

REG-VIRTUAL an online-only conference → a registration line, flagged

separate out

6 situations

one receipt becomes two lines, on two accounts

The Purple Pig

444 N Michigan Ave, Chicago, IL 60611

08/12/2026 1:12 PM Check #70637
Server: Camila Table: 6 Guests: 3

1	Pasta Primavera	22.00
1	Grilled Salmon	28.00
1	Fish Tacos	21.95
1	Old Fashioned	16.00
1	Old Fashioned	16.00
Subtotal		103.95
Sales Tax 10.25%		10.65
Restaurant Tax 0.5%		0.52
Tip		20.79
TOTAL USD		135.91

MEAL-BUSINESS-ALCOHOL lunch with two cocktails → 94.07 business meal with the attendees, 41.84 to entertainment (52611), flagged

file part

15 situations

the allowable part is filed; the full total must not appear

Hyatt Regency Chicago

101 Wacker Drive, Chicago, IL 60601

Guest Folio

Guest: Changhong Zhang Room: 1042 Folio: 0801807 Cost: PCH0807
Arrival: 08/10/2026 Departure: 08/14/2026 Rate plan: Conference rate — Standard King

Date	Description	Amount (USD)
08/10/2026	Room Charge	297.00
08/10/2026	Breakfast Operators Tax 6%	17.82
08/10/2026	Chicago Hotel Accommodations Tax 6.0%	13.97
08/10/2026	Cook County Hotel Tax 1%	2.97
08/10/2026	MPECA Hotel Tax 2.0%	7.43
08/10/2026	Sports Facilities Tax 2%	5.94
08/11/2026	Room Charge	253.00
08/11/2026	Breakfast Operators Tax 6%	15.18
08/11/2026	Chicago Hotel Accommodations Tax 6.0%	11.39
08/11/2026	Cook County Hotel Tax 1%	2.53
08/11/2026	MPECA Hotel Tax 2.0%	6.53
08/11/2026	Sports Facilities Tax 2%	5.06
08/12/2026	Room Charge	161.00
08/12/2026	Breakfast Operators Tax 6%	10.86
08/12/2026	Chicago Hotel Accommodations Tax 6.0%	8.15
08/12/2026	Cook County Hotel Tax 1%	1.81
08/12/2026	MPECA Hotel Tax 2.0%	4.83
08/12/2026	Sports Facilities Tax 2%	3.82
08/13/2026	Room Charge	229.00
08/13/2026	Breakfast Operators Tax 6%	14.30
08/13/2026	Chicago Hotel Accommodations Tax 6.0%	10.96
08/13/2026	Cook County Hotel Tax 1%	2.59
08/13/2026	MPECA Hotel Tax 2.0%	6.88
08/13/2026	Sports Facilities Tax 2%	4.70
08/14/2026	Room Charge	21.00
TOTAL USD 777.00 (Standard King)		21.00
Total		1160.88 USD
Payment — Visa 80004417		-1160.88 USD
Balance		0.00 USD

LOD-FOLIO-PERSONAL → 1,120.60 for room and tax, not the folio's 1,182.59

not filed

78 situations

no line at all, and the reason is said

jetBlue

Your Flight Receipt — JUN ZHANG — 24AUG26

Confirmation Number
E3HJAJ

Passenger Info
Name: JUN ZHANG

FLIGHT	DATE	DEPART	ARRIVE
JETBLUE 5198 Blue (B)	MON, 24AUG	RONALD REAGAN WASHINGTON NATIONAL AIRPORT 07:00AM	BOSTON LOGAN INTERNATIONAL AIRPORT 08:31AM

Flight Receipt

AIR-COMPANION the spouse's ticket on the traveler's booking → no line; the traveler's own ticket is filed

A FLAG HAS A LEVEL · NON-TRAVEL DATA

notice: filed as it is, the reader is told something — supplies paid out of pocket, with GW's preferred P-Card named (NC-SELF-PAID)

needs review: a person must decide — a computer under \$5,000 bought by the traveler, filed under 52193, since GW IT orders computers

JUDGMENT CALL

a situation whose answer is more than copying the receipt: every handling except “file”, 196 of 267. The answer key states the handling, so a score shows whether the agent judged right, not only whether it filled the form.

What the situations are about: 21 families of expense

Each situation belongs to one family, the kind of expense it is about: 10 families in travel and 11 outside it, so every kind of expense the policy decides has situations of its own.

TRAVEL · 169 SITUATIONS

UNITED

Confirmation Number

XPMPF2

Passenger Info

NAME: CHANGHONG ZHANG

FLIGHT

DATE

SEAT

ARRIVE

UNITED

18A

08/10/2026

12A

BOSTON/LOGAN/INTERNATIONAL

08/10/2026

08/10/2026

08/10/2026

AIR-COACH-DIRECT

a coach ticket

file

Hyatt Regency Chicago

Guest Folio

Guest: Changhong Zhang

Room: 1502

Folio: 9083107

Guest: PNC148807

Arrival: 08/10/2026

Departure: 08/14/2026

Rate plan: Conference rate - Standard King

Date

Description

Amount (\$USD)

08/10/2026

Room Charge

287.00

08/10/2026

Weekly Hotel Operators Tax 6%

17.62

08/10/2026

Chicago Hotel Administrative Tax 4.5%

13.27

LOD-FOLIO-PERSONAL

a folio with personal charges

file part

The Purple Pig

444 N Michigan Ave, Chicago, IL 60611

08/12/2026 1:12 PM

Check #70637

Server: Camila

Table: 6

Guests: 3

1

Pasta Primavera

22.00

1

Grilled Salmon

28.00

1

Fish Tacos

21.95

MEAL-BUSINESS-ALCOHOL

a business lunch with alcohol

separate out

Uber

Here's your receipt for your ride, changhong

August 25, 2026

Total

\$16.00

Tip

\$0.00

Booking Fee

\$0.00

Maximum Uber TNC Assessment

\$0.00

Payments

Apple Pay Visa ****4017

\$16.00

GT-UBER-TIP-UPDATED

a ride receipt replaced by an updated one

not filed

NEDS Society

THANK YOU!

Program Info

Date: 8/10/2026

Room Number: 90831070020

Customer: Changhong Zhang

Total

\$0.00

REG-VIRTUAL

an online-only conference

file + flag

Boston Public Golf Course

Receipt

Aug 26, 2026 9:12 AM

Pin: 1429409

Green Fee: 1.0

Cart

00.00

Total

\$10.00

OTH-CLUB-FEES

golf with collaborators

file + flag

NEDS 2026 - Northeast Data Science Symposium

Program at a glance

Tue Aug 26

09:00

Tuesday

10:00

Opening keynote

14:00

Sessions A-D

Wed Aug 27

09:00

Wednesday

09:00

Plenary session

14:00

Plenary session

Thu Aug 28

09:00

Thursday

09:00

Plenary session

14:00

Plenary session

DOC-PROGRAM

a conference program: not a receipt

not filed

foreign · 2

FX-BOTH-CURRENCIES

a card slip in the local currency and in dollars

file + flag

funding · 1

FUND-GRANT-ENTERTAINMENT

entertainment on a federal grant

not filed

timing · 1

TIME-REG-OLD-BUT-FINE

a registration paid 100 days before the trip

file

NON-TRAVEL · 98 SITUATIONS

REE

THANK YOU!

Receipt Info

Date: 8/10/2026

Receipt Number: 7636284964

Customer: Changhong Zhang

Total: \$258.00

QUANTITY

LINE DESCRIPTION

LIST PRICE

SALE PRICE

TOTAL

DUES-SOCIETY-ANNUAL

professional society dues

file + flag

Puget Systems

INVOICE

Bill to

Changhong Zhang

The George Washington University

Science & Engineering Hall, 800 22nd St NW, Washington, DC 20052

Invoice no.

89-019462

Invoice date

Aug 21, 2026

Terms

Due on receipt

Description

Qty

Unit price

Amount

Custom CPU workstation (Threackover P80, 2 x RTX 6000 Ada, 356 GB RAM)

1

\$6,450.00

\$6,450.00

SUPP-WORKSTATION-CAPITAL

a \$6,450 workstation: not in Concur

not filed

Allied Van Lines

INVOICE

Bill to

Changhong Zhang

1301 N Courthouse Rd, Arlington, VA 22201

Invoice no.

81L-748234

Invoice date

Aug 15, 2026

Terms

Due on delivery

Description

Qty

Unit price

Amount

Linehaul -- household goods shipment Chicago, IL to Arlington, VA

1

\$3,000.00

\$3,000.00

Packing service and materials

1

\$370.00

\$370.00

RELOC-FACULTY-MOVERS

movers for a new professor: paid through payroll

not filed

books, printing, postage, phone, other accounts · 17

BOOK-RESEARCH-TEXT

accounts the test account cannot select · 13

LEASE-COPIER-FIRST-PAYMENT

recruiting · 10

RECRUIT-CANDIDATE-DINNER

software · 7

SOFT-SAAS-ANNUAL

subscriptions · 6

SUBS-JOURNAL-ANNUAL

paid with the GW P-Card · 4

PCARD-SUBSCRIPTION

non-travel receipts in a trip report · 3

SUPP-ON-TRIP

books, printing, postage, phone, other accounts

accounts the test account cannot select

events, gifts, flowers

recruiting

software

subscriptions

paid with the GW P-Card

non-travel receipts in a trip report

file

not filed

file

file + flag

file + flag

file + flag

file + flag

file + flag

Appendix E2 ↗

Conditions, form work, traps, and the basis of every answer

Each situation also states where it applies, how much form work it takes, whether it hides a deliberate contradiction, and what its answer rests on.

CONDITIONS · WHERE IT APPLIES

city	US city 85 · abroad 18 · local, within 50 miles 9 · any 155 a lunch on a local business day: MEAL-LOCAL-PERSONAL not filed; the same lunch inside an all-day local meeting: MEAL-LOCAL-IN-MEETING file + flag
traveler	staff 22 · student 8 · faculty 13 · anyone 224 an Airbnb stay is flagged for staff (discouraged) and for a student without the advisor's written approval: LOD-AIRBNB-STAFF, LOD-AIRBNB-STUDENT
funding	federal grant 11 · not a grant 8 · any 248 hotel internet: filed on department funds, flagged on a grant (previous pages)

FORM WORK · HOW MUCH FILLING IT TAKES

simple · 204: one line from the receipt, AIR-COACH-DIRECT
complex · 63: more fields or a second screen, such as the folio's four nights, LOD-FOLIO-PERSONAL
Reported apart from the handling, so a score shows whether the agent filled the form badly or judged badly.

Recurrence ⓘ

Different daily amount

📅 Nights: 4

Date	Room Rate *	Room Tax	Tax 2	Tax 3
08/10/2026	297.00	47.53		
08/11/2026	253.00	40.49		
08/12/2026	181.00	28.97		
08/13/2026	235.00	37.61		

(Amounts in USD)Add Tax Fields

the folio's nights, as the agent entered them in Concur

TRAP · A DELIBERATE CONTRADICTION

19 travel situations hide one, to test whether the agent reads documents against each other.
AIR-WIFI-OTHER-CARRIER file + flag a Wi-Fi receipt from an airline the traveler did not fly: file the fee, flag the mismatch

American Airlines Wi-Fi (Viasat)

inflight wi-fi portal

Purchase Receipt

Aug 24, 2026 6:48 PM Ref: 31068415

Wi-Fi -- full flight pass, AA2406	10.00
Total	\$10.00
Paid: Visa ****4417	

Sample document generated for software testing (Expensify/benchmark, benchmark/pod). Not a real transaction, receipt or invoice.

a Wi-Fi pass on an airline other than the one the trip used

BASIS · WHAT THE ANSWER RESTS ON (SITUATIONS WITH EACH KIND)

- 228 a page of the manual or a GW rule · AIR-COMPANION: manual p.25, a companion only with the VP's approval
- 104 our decision, dated, where the policy is silent · LOD-AIRBNB-STUDENT
- 73 a source we looked up · DUES-SOCIETY-ANNUAL
- 64 experience, said as such: several answers are accepted · GT-TAXI
- 19 one of our own receipts · DOC-PROGRAM

Instance: generated documents with their answer key

An instance is one generated copy of a situation: one to three documents that look real, and their answer key, written by the same code.

THREE INSTANCES OF LOD-FOLIO-PERSONAL · EVERY SITUATION HAS THREE, 801 IN ALL

The Westin Copley Place, Boston

10 Huntington Ave, Boston, MA 02116 ·

Guest Folio

Guest: **Changhong Zhang** Room: 620

Arrival: 08/24/2026 Departure: 08/28/2026

Boston
Westin Copley Place
4 nights from 24 August

Palmer House, a Hilton Hotel

17 E Monroe St, Chicago, IL 60603 ·

Guest Folio

Guest: **Changhong Zhang** Room: 1436

Arrival: 07/13/2026 Departure: 07/16/2026

Chicago
Palmer House
3 nights from 13 July

Hyatt Regency Atlanta

265 Peachtree St NE, Atlanta, GA 30303 ·

Guest Folio

Guest: **Changhong Zhang** Room: 1977

Arrival: 09/14/2026 Departure: 09/19/2026

Atlanta
Hyatt Regency
5 nights from 14 September

What changes: the city, hotel, dates, nights, rates, taxes and the personal charges. **What stays:** the rule tested and the correct handling.

THE ANSWER KEY · WHAT THE SCORER CHECKS

```
// the expected line for the folio, trip A
"receipt": "Hyatt Regency Chicago folio.pdf",
"expense_type": "53104-US LODGING EXPENSE",
"amount": "1120.60", "currency": "USD",
"requires": itemization by night:
           08-10  297.00 + 47.53 tax
           08-11  253.00 + 40.49 tax
           08-12  181.00 + 28.97 tax
           08-13  235.00 + 37.61 tax
"unreduced_total": "1182.59" ← must not appear as a line
"situation": "LOD-FOLIO-PERSONAL"
```

WHAT AN ANSWER KEY CAN HOLD
the **lines** a report must contain: account, amount, date, vendor, currency, nights, attendees · the **flags** that must be raised, with the words they must use · the **documents to leave out**, with the reason · the **totals that must not appear** · what the agent's **summary** must mention

WHERE THE REALISM COMES FROM
12 cities with their real tax rules, airports and vendors · 6 airlines · 22 people · 11 conferences · 31 document layouts copied from real receipts

WHY IT CAN BE TRUSTED
amounts are computed to the cent before the document is drawn · seeded, so a rerun gives the same files · every PDF says it is a synthetic test document · 801 instances, 903 files

Trip: a story and its situations, filed as one expense report

A trip puts a traveler’s story and the situations it contains into one folder, which the agent files as one Concur report, a few documents per turn; the answer key stays hidden from the agent.

THE FOLDER · TRIP A, CHICAGO

the agent sees

· 26 documents and the notes

▶ Hyatt Regency Chicago folio.pdf

▶ Hyatt Regency Chicago reservation confirmation.pdf

· The Purple Pig business lunch check.pdf

· American Airlines DCA–ORD receipt.pdf

· American Airlines seat assignment receipt.pdf

· Uber Aug 11 receipt.pdf

· Uber Aug 11 UPDATED receipt.pdf

· Newsstand receipt.pdf

· Golf with the collaborators.pdf

· Conference program.pdf

· ... 16 more · trip_notes.txt

hidden from the agent

· expected.json – the answer key

· MANIFEST.md – the situations in the trip

· turns.json – the instructions, turn by turn

TRIP_NOTES.TXT, AS THE AGENT READS IT

Trip notes – MOSS 2026 Chicago – Changhong Zhang (postdoctoral researcher, GW School of Engineering)

Purpose: Present a paper at MOSS 2026 (Midwest Optimization & Systems Symposium), Chicago IL, Aug 11–13 2026, and meet the symposium's benchmark working group.

Dates: Aug 10 – Aug 14, 2026 in Chicago.

Filed on: Aug 19, 2026.

Funding: Non-Grant – departmental funds; the department will supply the Oracle Alias, use a placeholder for now.

Remarks and attendees:

– I moved my return flight on Aug 14 from the morning flight to the evening flight (fare difference \$108.00) for a follow-up meeting with the organizing committee that morning; my supervisor approved.

– Aug 12 lunch, The Purple Pig: Lena Hoffmann (University of Toronto), Ana Ferreira (McGill University), me. Benchmark paper planning.

The notes decide several documents: who sat at the Purple Pig lunch, why the return flight was changed.

THE TURNS · ONE RECORDED RUN EACH

turn 1 · 6 files: the flights, the seat, the bag, the changed return, the registration and the tutorial

turn 2 · 2 files: the hotel folio and the reservation confirmation

turn 3 · 7 files: the meals

turn 4 · 11 files: ground transport and the rest

Every turn begins: never submit; read trip_notes.txt first; file what the policy allows, leave out what it does not and say why, split what must be split, and flag every judgment call with its basis.

A TRIP MUST BE CONSISTENT

one way to travel and one ticket for the trip · one hotel at a time · fuel dated the day the car is returned · no two situations at one merchant on one day

23 TRIPS

A–O, 15 travel trips, use all 169 travel situations · **P–W**, 8 reports outside travel or mixed, use 94 of the 98 others · 382 documents, 257 lines, 134 documents not filed, 192 flags

PER DIEM

on a foreign leg, meals are paid as a daily allowance, so a meal receipt from that leg is not filed (**MEAL–INTL–PERDIEM**); trip E, New York and London, claims 6 days

Appendix E3 ➤

Scoring a filed report against its answer key

After a run we read the report back from Concur and check it against the key: trip A passed 84 of 87 checks (96.6%), its folio line passed every one, and the misses say exactly what went wrong.

WHAT THE AGENT FILED · TURN 2

Hyatt Regency Chicago

Chicago, Illinois

X

Payment Type *

Out of Pocket

Amount *

1,120.60

Currency *

US, Dollar (USD)

X

Comment

176/500

Folio \$1182.59 reduced by \$61.99: In-Room Movie \$14.99 (8/11), Minibar \$22.00 (8/12), Health Club \$25.00 (8/13) excluded as personal items. Conference rate, 4 nights Aug 10-14.

Nights: 4

Date	Room Rate *	Room Tax	Tax 2	Tax 3
08/10/2026	297.00	47.53		
08/11/2026	253.00	40.49		
08/12/2026	181.00	28.97		
08/13/2026	235.00	37.61		

(Amounts in USD)

Add Tax Fields

Concur after turn 2: the lodging line of 1120.60 with the deduction explained in its comment, and the four nights itemized

SCORE.TXT · TRIP A, BATCH 20260914-02

per-expense

...
✓ American Airlines DCA-ORD receipt.pdf 501.40 USD
✓ Hyatt Regency Chicago folio.pdf 1120.60 USD
✗ RPM Steak lunch check (large tip).pdf 70.61 USD MISSING
✗ Uber Aug 11 955 PM.pdf 20.10 USD MISSING

dimensions

coverage 21/23
expense_type 21/21
basic_fields 21/21
itemization 1/1
attendees 1/1
skip_respected 5/5
exclusions 1/1
flags_required 5/6
comments_in_concur 6/6
summary_mentions 2/2

△ FLAG MISSING: AIR-SEAT-ECONOMY – expected a flag mentioning one of ['...
△ 1 line(s) in the report match no expected line: 71.77 53102-US BUSINE...
OVERALL: 96.6% | submittable: True

THE TERMS

run: one turn, recorded step by step: the page seen, the decision, the actions, a screenshot. Trip A took 4 turns, 212 steps, \$7.22 and 24 minutes.

export: the report read back from Concur after the run.

score: checks passed ÷ checks in the key, reported by dimension.

not scored by machine: the wording of the business purpose, the Oracle alias, the mileage calculator's fields, whether an approval is attached; a person reads them in the trajectory.

ledger: each situation's record across batches: LOD-FOLIO-PERSONAL passed 3/3.

For learning across reports this gives, per report, a success score, the steps, the dollars and the seconds; and the same situations recur from trip to trip.

D · Evidence behind the five difficulties (1/2)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
WebCoach (Liu et al., 2025)	ICLR 2026 MemAgents workshop	External trajectory summary, additional Coach injects suggestions to action model	WebVoyager 643 real-time web tasks; dynamic library accumulates same-stream same-task but excludes same-task-ID retrieval	Skywork-38B + Qwen3-8B Coach: 0.614 vs 0.473; Qwen-VL-7B main table 0.311 vs 0.328	Skywork average per question 215 → 395 s, actions 10.7 → 10.2; time includes extra inference; no token cost measurement	● ○ ●
ReasoningBank (Ouyang et al., 2026)	ICLR 2026	Text experience pool extracted from success and failure trajectories	WebArena, SWE-bench; online single-task stream	WebArena Gemini-2.5-flash success rate 40.5 → 48.8; Shopping judge accuracy 72.7%	WebArena average ReAct steps 9.7 → 8.3; another table total tokens 50,847.4 → 53,054.5, baseline foundation not specified; time and cost not measured	● ○ ●
GenericAgent (Liang et al., 2026)	arXiv	SOP, script, hierarchical memory; fixed tool interface	Multiple models; local task, web, repeated download and research tasks	Same GPT-5.4 repeated download 102 seconds → ~66 seconds; not held-out; five long tasks success 100%, vs Claude Code 100%, OpenClaw 80%	Repeated download tokens 200,439 → 100,000; five long tasks 220.8 s vs OpenClaw 183.1 s; evolution cost not separated	● ● ●
EchoPath (Zhao et al., 2026)	arXiv	Parameterized GUI trajectory memory, precondition and lifecycle	OSWorld-Verified 159 task with established memory; same-task second try change resolution, codex-gpt-5.5-medium	91.2% vs Synapse 91.8%	Median tokens 20,370 vs 586,386; time 127.5 s vs 315.7 s; first build median ~572k tokens/4.5 min counted success-only	● ● ○
MobileGPT (Lee et al., 2024)	ACM MobiCom 2024	Per-app page-subtask graph and action sequence	Mobile apps; cold start then warm start same task with parameter change	GPT-4-turbo; with human repair ablation warm start 98.75%; main table without human repair cannot mix	Human-repair path warm start vs same prompt Derive-only: latency -62.5%, cost -68.8%; offline explore per app 10 to 15 minutes, total \$10.78	● ● ●
ActionEngine (Zhong et al., 2026)	arXiv	State-action graph, locator and descriptor, online general-purpose repair write-back	WebArena 655 tasks, Claude Opus 4.6; template warm-up then eval, patcher off	91.2%; no warm-up graph 73.1%	27 s vs 87 s; \$0.05 vs \$0.40, both per-task relative Claude Code; no explore/warm-up; break-even 39- 101 tasks and missing warm-up	● ● ●
Are Online Skill and Memory Modules Always Worth Their Tokens? (Hajimiri et al., 2026)	EMNLP 2026 main conference (listed in the programme)	Control AWM, ASI, ReasoningBank vs plain executor budget config	WebArena four domain, Gemini 3 Flash, three run	Vanilla-IB success 44.78%, 73.6K total tokens/question, that model four-config best simultaneously	Token only; plain 15 step+pruning, enhance 10 step, near-equal budget	● ● ●
ClawTrace / CostCraft (Yuan et al., 2026)	ACM CAIS 2026 Agent Skills workshop, non- archival	Trajectory-summary exported skill document, frozen model	SpreadsheetBench 30 held-out, OpenClaw + gpt-5.4; also migrate SkillsBench	Remove delete patch after two-type sum degrade 5 → 14; migrate post cost median not decrease	Execution cost with cache; SkillsBench median \$0.143 vs \$0.144; extract cost not reported	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ○ partly, ○ no (initial judgment, calc.; Appendix A1).

D · Evidence behind the five difficulties (2/2)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
SEDM (Xu et al., 2025)	NeurIPS 2025 SEA workshop	Injectable memory fragments with utility weight and source	LoCoMo, FEVER, HotpotQA; independent build vs evaluation not clear	GPT-4o-mini FEVER: 66 vs G-Memory 62, no memory 57; HotpotQA 39 / 38 / 34	FEVER prompt tokens 2.47M vs G-Memory 3.62M, no memory 1.65M; replay cost and time cost not measured	● ○ ●
HarnessFix (Chen et al., 2026)	arXiv	Prompts, tool description, orchestration, config, adapter and validation script	GAIA, SWE-bench Verified, AppWorld, Terminal-Bench 2.0 Verified; GPT-5 mini	Initial system to patched completion +6.3 to +18.4 percentage points; mean of three runs	AppWorld offline patch 37.2 million tokens; execution-phase time/cost not reported	○ ● ●
ESPO (L. Liu et al., 2026)	EMNLP 2026 Findings (listed in the programme)	Natural-language module prompt	Seven text benchmark, each 500 held-out; Claude Sonnet 4.5	Main optimize single average 74.67% vs GEPA 70.91%; prompt 1,004 vs 1,878 character	PUPA optimize ~3.4M tokens, 2h 13m; no deployment acceleration measured	○ ● ●
GEPA (Agrawal et al., 2026)	ICLR 2026	Module prompts only, model frozen	QA, instructions, privacy rewriting, math; train validation test separated	Qwen3 8B AIME-2025: 32.00 vs MIPROv2 20.00, GRPO 38.00; overall advantage ≠ every task wins	Per-task optimization rollout total 6871 / 3593 / 7051 / 2426 / 1839 / 1839; GRPO per item 24000; not model calls, time or cost	○ ● ●
SICA (Robeyns et al., 2025)	ICLR 2025 Self-Improving Foundation Models Without Human Supervision workshop	Self-complete Python code, tools and oversight mechanism	Sonnet3.5(v2), o3-mini; fixed coding subset and synthetic-file task	SWE peak 17% → 53%, final accuracy 0.51; same baseline for agent selection, not held-out	Cross-four-benchmark full-attempt per-question average: final \$1.91 → \$1.70, 130.2 → 114.5 s; tokens 0.24 → 0.30 M; full cycle ~\$7,000	● ● ○
ADAS (Hu et al., 2025)	ICLR 2025 main conference	Fixed small framework forward() program	GPT-4o search, GPT-3.5 execution; ARC, QA reasoning held-out test	DROP F1 79.4 vs 65.8; MGSM 53.4 vs 39.0, each strongest hand-written baseline	One search + eval ~500 USD (ARC), 300 USD (reasoning); deployment per-question not measured	○ ● ●
Group-Evolving Agents / GEA (Weng et al., 2026)	arXiv	Multiple parent agents share trajectories then modify framework code	SWE-bench/Polyglot; Haiku4.5 then Sonnet4.5, o1 reflection	Report SWE 71.0%, Polyglot 88.3%, but latter inconsistent with claimed 50-question held-out denominator; not use as generalization-gain evidence	Full cycle estimate USD 13,000 / USD 1,500; no per-question post-evolution cost change	○ ● ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ○ partly, ○ no (initial judgment, calc.; Appendix A1).